DATABASE MANAGEMENT SYSTEMS

Design And Implementation Of A DBMS For Retail Inventory & Sales

PROJECT-BASED LEARNING REVIEW 1

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PROBLEM STATEMENT

The Challenge

Retail stores handle massive volumes of data related to products, suppliers, purchases, sales, and stock movements.

Managing these records manually or using disconnected flat files consistently leads to critical business errors:

- Negative stock balances
- Inconsistent invoicing and pricing
- Difficulty tracking customer purchases
- Poor handling of product returns

The Solution

The **Retail Inventory & Sales Management System** provides a centralized, relational database architecture designed to manage the entire retail lifecycle.

By migrating to a structured DBMS, the store can operate efficiently while maintaining absolute data integrity, ensuring real-time inventory tracking, and establishing a robust foundation for future application development.

SYSTEM SCOPE & OBJECTIVES

System Scope

The system manages the core retail operations of a modern store, encompassing both the procurement and fulfillment lifecycles.

Major Entities Tracked:

PRODUCT | CATEGORY | BRAND | SUPPLIER | PURCHASE | PURCHASE_ITEM | CUSTOMER | SALE | SALE_ITEM | RETURN | STOCK_MOVEMENT | INVOICE | PAYMENT

Key Objectives

- Design a structured relational database for retail operations.
- Eliminate redundant and inconsistent data storage.
- Maintain accurate, real-time inventory information.
- Establish clear relationships between core business entities.
- Enforce strict data integrity using primary/foreign key constraints.
- Track purchases, sales, returns, and stock movements accurately.

RELATIONAL SCHEMA (1/2)

product_id
product_code
product_name
category_id FK
brand_id FK
supplier_id FK
unit_price
stock_quantity

purchase_id
supplier_id FK
purchase_date
total_amount

purchase_item_id
purchase_id FK
product_id FK
quantity
unit_cost

category_id
category_name

brand_id
brand_name

supplier_id
supplier_name
contact_number

THANK YOU

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